

Lower Expected Heating Costs or Less Exposure to Price Spikes: What Matters Most for Heat-Pump Adoption?

Pre-Analysis Plan

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September 23, 2026

Abstract

This study examines whether households' willingness to adopt an air-to-water heat pump is shaped more by its expected operating-cost advantage or by its lower exposure to energy-price shocks, relative to a conventional gas boiler. A randomized information experiment separately varies information about the expected effect of ETS 2 on future gas-heating costs and about the declining transmission of gas-price shocks to electricity prices. Beliefs are elicited about the distribution of future monthly heating costs and about the probability of unusually high monthly costs for both technologies. The experiment enables estimation of the causal effects of relative expected heating costs and relative exposure to high heating costs on heat-pump adoption intentions, as well as comparison of the importance of these two channels.

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1 Motivation

Climate change and energy sovereignty are among the major challenges facing Europe today. Electrification is widely seen as one of the main levers for decarbonizing economies while reducing dependence on fossil fuel imports and exposure to geopolitical shocks. Recent events, such as the war in Ukraine and tensions in the Middle East, have highlighted this vulnerability by significantly increasing household energy bills across Europe.

Whether households contribute to the electrification of residential heating ultimately depends on their willingness to replace fossil-fuel heating systems with heat pumps. Heat-pump adoption is shaped by financial determinants, dwelling characteristics, and non-financial considerations such as comfort, environmental attitudes, familiarity with the technology, and access to qualified installers (Davis, 2024; Lang et al., 2021; Meles et al., 2022). Among the financial determinants, expectations about future electricity costs relative to those of conventional fossil fuels (primarily natural gas and heating oil) are particularly important, as they shape both the long-term profitability and the financial predictability of heat-pump investments (Alberini et al., 2013).

Two developments are set to reshape both the long-term profitability and the financial predictability of heat-pump investments. First, the new European Emissions Trading System (ETS 2), scheduled to be introduced in 2028, will impose a carbon price on fossil fuels used for residential heating. Although ETS 2 regulates fuel suppliers rather than households, its cost is expected to be fully passed through to final consumers (Andersson, 2019). At an assumed introductory price of €60/t CO₂ (Federal Planning Bureau, 2025), this would increase the monthly bill of an average Belgian household heating with natural gas by about €18 per month¹.

Second, the expansion of renewable generation and flexibility resources is expected to weaken the sensitivity of electricity prices to unpredictable shocks, including geopolitical disruptions. European wholesale electricity markets largely rely on marginal pricing: generators are dispatched in increasing order of marginal cost and the market-clearing price is determined by the last unit required to meet demand. Because gas-fired power plants have frequently been marginal, European electricity prices have been strongly exposed to natural-gas price fluctuations, despite gas accounting for a much smaller share of electricity generation (Zakeri et al., 2023). This exposure became particularly salient during the energy crisis triggered by Russia's

¹ This estimate combines CREG's standard residential gas consumption of 17,000 kWh per year with the EU ETS emission factor for natural gas of 56.1 t CO₂/TJ (0.202 kg CO₂/kWh), implying €17.2 per month at €60/t CO₂ (Commission de Régulation de l'Électricité et du Gaz, 2022; European Commission, 2018).

invasion of Ukraine. In the coming years, the expansion of wind and solar generation, together with batteries, interconnection, and demand-side flexibility, is expected to reduce the frequency with which gas-fired generation sets the marginal price and to weaken the sensitivity of electricity prices to gas-price shocks (Navia Simon & Diaz Anadon, 2025). Their simulations indicate that Belgian annual electricity prices will become markedly less sensitive to natural-gas prices by 2030, while ICIS, a provider of energy-market data, analysis and forecasts, projects a full annual-average decoupling by 2032 (Jackson-Stroud, 2026).

These developments highlight two distinct economic channels through which future energy prices may affect heat-pump adoption. First, an increase in expected gas-heating costs relative to heat-pump operating costs makes heat pumps more financially attractive compared with gas boilers. Empirical studies show that heat-pump adoption is higher where natural gas is relatively expensive and electricity relatively cheap (Anderson & Kirkpatrick, 2024; Davis, 2024; Meier & Lehmann, 2026; Sahari, 2019). Second, the relative uncertainty surrounding future prices may matter independently of their expected levels. A technology whose operating costs are less exposed to future upward price shocks can provide households with a hedge against energy-price risk. Although comparable evidence for heat-pump installations is lacking, studies of vehicle choice show that uncertainty about future operating costs affects investment preference (Archsmith & Levin, 2026; Ding et al., 2025).

No previous study appears to have examined whether households place greater value on lower expected average costs or on protection against upward energy-price shocks when deciding whether to invest in a heat pump. The closest experimental evidence is provided by Eßer et al. (2024), who study the impact of lifetime-cost information and price-range information on German households' stated intentions to invest in heat pumps. Their Lifetime-cost information treatment, indicating that the heat pump is cheaper, increases willingness to pay relative to the control group. Their second treatment adds information portraying future gas prices as more uncertain than electricity prices. The combined treatment is associated with higher willingness to pay for the heat pump than the lifetime-cost treatment alone, but the difference is not statistically significant. However, the experiment does not cleanly identify the two channels. Indeed, as emphasized by Bloom (2014), changes in expected outcomes and uncertainty are often interconnected. The price-range treatment is intended to provide information about the second moment of beliefs but may also shift the first moment, namely expected relative operating costs, because the projected gas-price range is less favourable than the electricity-price

range. Conversely, the lifetime-cost treatment, although primarily providing information about expected relative costs, may also affect households' perceived exposure to future price shocks. Since the treatments do not independently vary the two dimensions, the authors cannot identify their separate effects and therefore cannot determine whether households place greater value on lower expected average costs or on protection against upward price shocks.

Identifying the causal effects of these two channels is therefore challenging. In their study of macroeconomic uncertainty and household spending, Coibion et al. (2024) address this identification problem by randomly assigning households to information treatments designed to shift the first moment, the second moment, or both moments of expectations about future economic growth. Post-treatment belief measures reveal how each intervention affects the two moments. These differential first-stage effects provide instruments for households' expected growth and uncertainty, allowing the authors to estimate the causal effect of each moment on spending. Related experiments use this approach to study inflation expectations, uncertainty, and household decisions (Fischer et al., 2024; Georgarakos et al., 2024; Kostyshyna & Petersen, 2024).

I apply this identification strategy to residential heating. The main objective is to compare the importance of relative heating-cost expectations and relative exposure to high heating costs in households' decisions to switch from fossil-fuel heating to a heat pump, and to determine which mechanism matters more. The key challenge is that both relative expected costs and relative exposure may be endogenous. For example, households that are already more willing to adopt a heat pump may pay more attention to future energy prices and therefore report both different expectations and lower perceived exposure to high costs. This would create reverse causality. In addition, unobserved environmental preferences may make households both more willing to adopt a heat pump and more likely to hold beliefs favourable to heat pumps. For example, they may expect climate policy to make gas heating relatively more expensive than heat-pump heating, or expect the expansion of renewable electricity to reduce electricity prices' exposure to fossil-fuel shocks. This creates omitted-variable bias. These sources of endogeneity prevent a causal interpretation of simple correlations. To address this issue, the study implements a randomized information experiment that separately varies information about expected relative heating costs and about the relative exposure of natural-gas and electricity prices to future shocks. The relative-cost treatment informs respondents about the expected effect of ETS 2 on the future cost of natural-gas heating relative to electric heating. The shock-exposure treatment provides information about how renewable generation, battery storage, and other

sources of flexibility are expected to reduce the exposure of electricity prices to fossil-fuel price shocks. A combined treatment provides both pieces of information. Respondents' subjective expectations and uncertainty are elicited before and after treatment to verify how each intervention changes both belief measures, and the exogenous variation generated by the experiment is used to estimate and compare their causal effects on households' willingness to invest in a heat pump.

2 Experimental Design

2.1 Pre-treatment questions

2.1.1 Tenure and Current Heating System

Respondents are first asked whether they own or rent their home and which heating system is currently used: a gas boiler, an oil-fired boiler, a heat pump, a pellet boiler, or electric heating.

All respondents are then asked how likely they think it is that their household will move from its current home within the next 10 years, using a scale from 0 (not at all likely) to 10 (extremely likely).

2.1.2 Perceptions of Heat-Pump Purchase Costs

Respondents are informed that replacing the heating system they report costs approximately a corresponding average amount in Belgium. They are then given a short, non-technical explanation of an air-to-water heat pump, emphasizing that it uses electricity to extract heat from outside air, heats water circulated through the home's radiators, and can also provide hot water. Respondents estimate the average purchase and installation cost of an air-to-water heat pump.

2.1.3 Initial Intention to Adopt a Heat Pump

Before treatment assignment, respondents report on a five-point scale whether they would consider installing an air-to-water heat pump when their current heating system next needs to be replaced.² The response categories range from "definitely not" to "definitely yes." This provides a pre-treatment measure of adoption intentions without presenting a specific heat-pump price.

² For respondents who rent their home, every personal heating-system adoption question additionally includes the following instruction: "Please imagine that you would have to pay the full purchase and installation costs yourself and that you would remain living in your current home for the next 20 years."

2.1.4 Perceptions about Current and Future Heating Costs

Respondents are shown the average monthly heating costs of a typical Belgian home with both a conventional gas boiler and an air-to-water heat pump during the first half of 2021, when natural-gas and electricity prices were at their normal levels. They are then told that Russia’s invasion of Ukraine triggered an energy crisis that had a major impact on energy prices in Belgium, and they estimate the average monthly heating cost of the same home with each technology during the first half of 2022. This provides a common reference point for normal costs of both systems while eliciting respondents’ prior beliefs about the extent to which the energy shock was transmitted from gas to electricity.

Before treatment, respondents provide their expected average monthly heating cost in 2030 for both technologies. For each technology, they then report the probability that the monthly cost in 2030 will be at least 20% higher than their expected average. The corresponding threshold, equal to 120% of the respondent’s reported average, is calculated and displayed by the survey software. These measures provide baseline beliefs that correspond directly to the post-treatment relative-cost and high-cost-exposure measures.

2.1.5 Socio-Demographics

Before treatment assignment, the survey also collects household monthly income, age, education, region of residence, and household size.

2.1.6 Risk Preferences

Before treatment assignment, respondents complete a short hypothetical lottery-choice task adapted from the simple gamble-choice approach of Eckel and Grossman (2008). They choose one of six options. Each option gives a 50% probability of receiving a lower amount and a 50% probability of receiving a higher amount. Option A pays 28 euros with certainty. Moving from Option B to Option E, both the expected payoff and the dispersion of possible payoffs increase; Option F has the same expected payoff as Option E but greater dispersion. Respondents are explicitly informed that no real payment will be made and are asked to answer as if their selected lottery would actually be played and paid to them. Choices of safer lotteries are interpreted as indicating greater risk aversion.

2.2 Treatment

2.2.1 Control Group

Subjects in the control group do not receive any additional information and move directly from the pre-treatment questions to the post-treatment questions. This group provides the benchmark against which treatment effects are measured.

2.2.2 Treatment 1

Subjects in the first treatment group receive information about expected relative heating costs:

At present, the monthly energy costs of heating an average Belgian home with a gas boiler or an air-to-water heat pump are broadly comparable.

Following the introduction in 2028 of a European carbon tax on fossil fuels used for residential heating, heating the home with gas is expected to cost approximately 50% more per month than heating it with an air-to-water heat pump by 2030.³

2.2.3 Treatment 2

Subjects in the second treatment group receive information about the expected transmission of natural-gas price shocks to electricity prices:

Today, natural-gas and electricity prices tend to move together. This is because gas-fired power plants are often needed to meet electricity demand. As a result, they set the wholesale electricity price.

As renewable electricity generation and battery storage expand, gas-fired power plants will be needed less often to meet electricity demand. As a result, from 2030 onwards, natural-gas price shocks are expected to have only half their current effect on Belgian electricity prices, and this link is expected to weaken further over time.⁴

For illustration, if a gas-price shock caused electricity prices to increase by €20 today, the same shock would cause an increase of around €10 in 2030.

³ This illustrative difference is calibrated using an internal calculation from the Renobel project dossier. For a Belgian household in a PEB-C dwelling with a current natural-gas heating bill of approximately €100 per month, a carbon price of €100/tCO₂ adds about €19 per month to the gas bill. Assuming a seasonal coefficient of performance of 3 for the air-to-water heat pump and an electricity price of €0.30/kWh, the calculation yields monthly heating costs of approximately €119 with gas (including the carbon charge) and €82 with the heat pump. This comparison is used to motivate the rounded 50% cost difference in the treatment.

⁴ This projection draws on simulations by Navia Simon and Diaz Anadon (2025). In their reference scenario, a €1 increase in natural-gas prices raises annual electricity prices by about €1.40; by 2030, the corresponding increase is about €1.00. Under stronger renewable deployment, the increase could fall below €0.50. The treatment translates this evidence into a simplified statement for respondents.

2.2.4 Treatment 3

Subjects in the third treatment group receive both Treatment 1 and Treatment 2.

2.3 Post-Treatment questions

2.3.1 Outcomes (1): Heating-Cost Expectations and Relative Exposure to High Heating Costs

In Question 15, respondents’ expectations about future heating costs for both a conventional gas boiler and an air-to-water heat pump are re-elicited using an interactive distribution interface. Respondents provide, separately for each technology, their subjective distribution by assigning probabilities to monthly cost levels ranging from 30 euros to 250 euros in 10-euro increments. When they have finished, they click a “Normalize” button, which rescales the probabilities to sum to 100% and displays the probability-weighted average cost implied by the respondent’s distribution for each technology. For each technology, the post-treatment probability of a high monthly cost is then recovered directly from this respondent-provided distribution: it is the cumulative probability assigned to all monthly cost levels at or above 120% of the corresponding probability-weighted average cost.

Respondents then report how many months between January 2030 and December 2035 they expect a major energy-price shock to cause the monthly heating cost to reach or exceed 1.2 times the calculated average cost for the relevant technology. This individualized threshold is displayed in euros for each heating system. The number must be between 0 and 72, and the months need not be consecutive.

Questions 15 and 16 are deliberately complementary. The distribution elicited in Question 15 measures whether respondents incorporate the information treatment into their overall subjective distribution of future heating costs without explicitly directing their attention to energy-price shocks. Question 16 instead measures beliefs about high heating costs conditional on a major energy-price shock and is therefore more directly aligned with the mechanism described in Treatment 2. The distribution-based measure from Question 15 is the main measure of relative exposure to high heating costs, while the shock-conditioned months measure from Question 16 is a pre-specified alternative used in the robustness analysis. If Treatment 2 produces only a weak shift in the distribution-based measure but a clear shift in the shock-conditioned measure, this pattern will be interpreted as evidence that the treatment changes beliefs about heating costs conditional on a shock without clearly changing respondents’ overall subjective cost distributions. Both analyses will be reported rather than selecting between the measures based on their statistical significance.

2.3.2 Outcomes (2): Heat-Pump Adoption

- Respondents are asked to imagine that their current heating system needs to be replaced and to report the maximum purchase and installation price they would be willing to pay for an air-to-water heat pump, given that a conventional gas boiler costs 5,000 euros on average.
- Respondents then rate, on a scale from 0 (not at all likely to choose) to 10 (extremely likely to choose), how likely they would be to choose a conventional gas boiler, a pellet boiler, an oil-fired boiler, and an air-to-water heat pump when replacing their current heating system, taking the stated average purchase and installation costs into account.
- Finally, respondents are informed that 2% of homes in Belgium currently have a heat pump and are asked to predict this share in 2030.

3 Setting and Sample Size

The experiment will be conducted with a sample of respondents representative of the Belgian adult population. Recruitment will ensure that the composition of the sample reflects the main socio-demographic characteristics of the population. The unit of randomization is the individual respondent. After completing all pre-treatment questions, respondents are randomly assigned by the survey software with equal probability to the control group, Treatment 1, Treatment 2, or Treatment 3. Randomization is implemented independently across respondents without blocking or stratification.

The study has five main outcomes: the two belief outcomes, $RelExp_i$ and $RelShock_i$, and the three heat-pump-adoption outcomes—the 0–10 likelihood rating, willingness to pay, and the expected 2030 adoption share. The required sample size is based on the most restrictive of these five outcomes. The power analysis uses a two-sided significance level of 5% and a target power of 80%. The calculation assumes that the most restrictive outcome requires detecting an effect of 0.15 standard deviations. For a comparison between one treatment arm and the control group, with equal allocation across the four experimental groups, this requires approximately 2,800 analyzable observations, or 700 respondents per group. The planned regressions control for the corresponding pre-treatment measures, which should improve the precision of the intention-to-treat estimates, although this gain is not quantified in the present calculation.

The target of 2,800 analyzable observations is therefore justified for the pairwise intention-to-treat comparisons, but it does not establish 80% power for the primary 2SLS comparison

between the two belief coefficients. The precision of that comparison also depends on the strength and differential relevance of the instruments for the two endogenous beliefs, as well as on the covariance between the two estimated coefficients. For planning purposes, the calculation assumes that the precision gains from controlling for the corresponding pre-treatment outcomes and other prognostic pre-treatment covariates will compensate for the loss of precision associated with instrumental-variables estimation and the inclusion of treatment-by-prior interactions. On this basis, the target of 2,800 analyzable observations is retained.

4 Main hypotheses and Mechanisms

4.1 Hypotheses on Beliefs

- Respondents with a larger gap between their prior beliefs and the information provided about 2030 heating costs are expected to update their expected relative-cost beliefs more strongly after treatment.
- The shock-exposure treatment is expected to increase respondents' perceived probability of high heating costs for gas heating relative to heat-pump heating. This effect should be larger among respondents with a lower pre-treatment value of $RelShock_i$, who have more scope to revise their beliefs towards a greater relative exposure of gas heating.

4.2 Hypotheses on Heat-Pump Adoption

- Individuals with a higher perceived ratio of gas-heating to heat-pump expected costs, $RelExp_i$, are expected to be more inclined to invest in a heat pump.
- Individuals who perceive gas heating as relatively more exposed to high heating costs than heat-pump heating, as captured by a higher value of $RelShock_i$, are expected to be more inclined to invest in a heat pump.
- The shock-exposure treatment is expected to increase heat-pump adoption more strongly among more risk-averse respondents. Because these respondents place greater value on protection against unexpectedly high heating costs, the reduction in the perceived exposure of heat-pump operating costs to energy-price shocks should be more attractive to them.

5 Analysis

5.1 Baseline Balance

Baseline balance across treatment groups will be assessed using pre-treatment covariates, including gender, age, income, education, employment status, expected residential mobility, and risk aversion. For each covariate, regressions of the covariate on treatment-group indicators will be estimated, followed by a test of whether the treatment indicators are jointly significant. For categorical covariates, balance will be tested using category indicators or global tests of equality in distributions across treatment groups.

5.2 Belief Measures

5.2.1 Relative Expected Heating Cost

Relative expected heating cost is defined as the ratio between the expected operating cost of the natural-gas technology and that of the heat pump:

$$RelExp_i = \frac{E_i(C_{2030}^{gas})}{E_i(C_{2030}^{HP})} \quad (5.2.1)$$

where $E_i(C_{2030}^{gas})$ denotes respondent i 's expected heating cost with a natural-gas system in 2030, and $E_i(C_{2030}^{HP})$ the corresponding expected heating cost with a heat pump. A value of one means that the two expected costs are equal. A value above one means that the respondent expects gas heating to cost more than heat-pump heating, while a value below one means the reverse.

5.2.2 Relative Exposure to High Heating Costs

For each technology $j \in \{gas, HP\}$, define the subjective probability of a high monthly heating cost in 2030 as

$$HighProb_i^j = P_i \left(C_{2030}^j \geq 1.2 E_i(C_{2030}^j) \right), \quad (5.2.2)$$

where probabilities are expressed on a zero-to-one scale. **Relative exposure to high heating costs** is defined as the difference between the gas and heat-pump probabilities, expressed in percentage points:

$$RelShock_i = 100 \left(HighProb_i^{gas} - HighProb_i^{HP} \right). \quad (5.2.3)$$

Before treatment, $HighProb_i^j$ is elicited directly after the survey software displays the

threshold equal to 120% of the respondent's reported average cost for technology j . After treatment, it is calculated from the elicited distribution as the cumulative probability assigned to monthly cost levels at or above 120% of the probability-weighted average for that technology. A positive value of $RelShock_i$ means that the respondent assigns a higher probability of unusually high costs to gas heating than to the heat pump; a negative value means the reverse.

5.2.3 Standardization

As in Settele and Shupe (2022), each post-treatment belief measure is standardized using the control-group mean and standard deviation to express expected relative costs and relative exposure to high heating costs on comparable scales.

The notation $RelExp_i$ and $RelShock_i$ is used for the post-treatment beliefs in their natural units, with standardized measures defined as

$$RelExp_i^{std} = \frac{RelExp_i - \overline{RelExp}_C}{s_C(RelExp)}, \quad RelShock_i^{std} = \frac{RelShock_i - \overline{RelShock}_C}{s_C(RelShock)}. \quad (5.2.4)$$

Here, the subscript C denotes the control group. The terms $\overline{RelExp}_C$ and $\overline{RelShock}_C$ denote the control-group means, while $s_C(RelExp)$ and $s_C(RelShock)$ denote the corresponding control-group standard deviations. The superscript std is used throughout the main treatment-effect and 2SLS equations whenever a standardized post-treatment belief enters the model. By contrast, $RelExp_i^{pre}$, $RelShock_i^{pre}$, and $GapCost_i^{pre}$ remain in their natural units: a ratio, a percentage-point difference, and a ratio-point gap, respectively. The adoption outcomes are not standardized.

5.3 Treatment effect on Beliefs

To estimate whether the information treatments successfully shift beliefs, separate specifications are estimated for relative expected heating costs and relative exposure to high heating costs. In both specifications, T_{1i} , T_{2i} , and T_{3i} are indicators for the three treatment groups, while the control group is the reference category. The vector $\mathbf{X}_i$ contains the socio-demographic and dwelling controls.

5.3.1 Effect on Relative Expected Heating Cost

This specification estimates whether the information treatments change relative expected heating costs and whether their effects vary with respondents' initial beliefs relative to the Treatment 1 signal. The outcome, $RelExp_i^{std}$, is the standardized post-treatment version of the natural-unit ratio $RelExp_i$ defined in Section 5.2.1. The baseline measure, $RelExp_i^{pre}$, is the

corresponding pre-treatment ratio between respondents' 2030 average monthly cost estimates for gas heating and heat-pump heating.

To capture respondents' initial distance from the information provided, the pre-treatment information gap, $GapCost_i^{pre}$, is defined as the signed difference between this baseline ratio and the gas-to-heat-pump cost ratio of 1.5 stated in Treatment 1:

$$GapCost_i^{pre} = RelExp_i^{pre} - 1.5.$$

This gap enters the regression through an interaction with each treatment indicator. The following specification is estimated:

$$\begin{aligned} RelExp_i^{std} = & \alpha + \rho RelExp_i^{pre} + \sum_{k=1}^3 \beta_k T_{ki} \\ & + \sum_{k=1}^3 \lambda_k (T_{ki} \times GapCost_i^{pre}) + \gamma' \mathbf{X}_i + \varepsilon_i. \end{aligned} \quad (5.3.1)$$

Following Coibion et al. (2024), these interactions allow the slope relating prior and posterior beliefs to differ across all treatment arms. For Treatments 1 and 3, respondents with a prior ratio below 1.5 have a negative gap and are expected to revise $RelExp_i$ upwards, whereas respondents with a prior ratio above 1.5 have a positive gap and are expected to revise it downwards. The magnitude of updating is expected to increase with $|GapCost_i^{pre}|$.

5.3.2 Effect on Relative Exposure to High Heating Costs

This specification estimates whether the information treatments change respondents' perceived probability of high heating costs for gas heating relative to heat-pump heating, and whether these effects vary with their initial beliefs. The outcome, $RelShock_i^{std}$, is the standardized post-treatment version of the natural-unit percentage-point difference $RelShock_i$ defined in Section 5.2.2, while $RelShock_i^{pre}$ is the corresponding pre-treatment difference in percentage points. The following specification is estimated:

$$\begin{aligned} RelShock_i^{std} = & \alpha + \rho RelShock_i^{pre} + \sum_{k=1}^3 \beta_k T_{ki} \\ & + \sum_{k=1}^3 \lambda_k (T_{ki} \times RelShock_i^{pre}) + \gamma' \mathbf{X}_i + \varepsilon_i. \end{aligned} \quad (5.3.2)$$

The treatment-specific interactions allow the slope relating prior and posterior beliefs to differ across all three treatment arms.

5.4 Heat-Pump Adoption

Following Coibion et al. (2024), a two-stage least squares (2SLS) model is estimated in which the two standardized post-treatment beliefs, $RelExp_i^{std}$ and $RelShock_i^{std}$, are treated as endogenous. The excluded instruments are the three treatment indicators and, for each treatment arm j , its separate interactions with both pre-treatment beliefs: $T_{ji} \times GapCost_i^{pre}$ and $T_{ji} \times RelShock_i^{pre}$.

The first-stage equations (5.4.1) and (5.4.2) and the second-stage equation (5.4.3) are as follows:

$$\begin{aligned} RelExp_i^{std} = & a_0 + \sum_{j=1}^3 a_j T_{ji} + \rho_1 RelExp_i^{pre} + \rho_2 RelShock_i^{pre} + \delta_E Adopt_i^{pre} \\ & + \sum_{j=1}^3 b_j (T_{ji} \times GapCost_i^{pre}) + \sum_{j=1}^3 c_j (T_{ji} \times RelShock_i^{pre}) \\ & + \gamma' \mathbf{X}_i + u_i \end{aligned} \quad (5.4.1)$$

$$\begin{aligned} RelShock_i^{std} = & \tilde{a}_0 + \sum_{j=1}^3 \tilde{a}_j T_{ji} + \tilde{\rho}_1 RelExp_i^{pre} + \tilde{\rho}_2 RelShock_i^{pre} + \tilde{\delta}_S Adopt_i^{pre} \\ & + \sum_{j=1}^3 \tilde{b}_j (T_{ji} \times GapCost_i^{pre}) + \sum_{j=1}^3 \tilde{c}_j (T_{ji} \times RelShock_i^{pre}) \\ & + \gamma' \mathbf{X}_i + v_i \end{aligned} \quad (5.4.2)$$

$$\begin{aligned} HeatPump_i = & \alpha + \theta_1 \widehat{RelExp}_i^{std} + \theta_2 \widehat{RelShock}_i^{std} \\ & + \rho_1 RelExp_i^{pre} + \rho_2 RelShock_i^{pre} + \delta Adopt_i^{pre} + \gamma' \mathbf{X}_i + \varepsilon_i \end{aligned} \quad (5.4.3)$$

where:

- $RelExp_i^{std}$ is the standardized post-treatment ratio of expected gas-heating costs to expected heat-pump heating costs.
- T_{ji} indicates assignment to treatment arm $j \in \{1, 2, 3\}$, while the control group is the reference category.
- $RelExp_i^{pre}$ and $RelShock_i^{pre}$ are the pre-treatment gas-to-heat-pump expected-cost ratio and the difference in perceived high-cost probabilities between gas and heat-pump heating, respectively. They remain in their natural units: a ratio and a percentage-point difference.

- $Adopt_i^{pre}$ is the respondent's pre-treatment intention to adopt an air-to-water heat pump, measured on the 1–5 scale in Question 5. It is included as a baseline adoption-intention control in all three outcome-specific models.
- $GapCost_i^{pre} = RelExp_i^{pre} - 1.5$ is the signed difference between the respondent's pre-treatment expected-cost ratio and the ratio stated in Treatment 1.
- $\mathbf{X}_i$ is the vector of socio-demographic and dwelling controls included in all three equations.
- u_i is the error term in the expected-cost first-stage equation.
- $RelShock_i^{std}$ is the standardized post-treatment difference in perceived high-cost probabilities between gas and heat-pump heating.
- v_i is the error term in the high-cost-probability first-stage equation.
- $HeatPump_i$ denotes one of three complementary heat-pump outcomes: the 0–10 likelihood rating assigned to a heat pump in the stated-choice task, the maximum purchase and installation price the respondent is willing to pay for a heat pump rather than a gas boiler, and the expected share of gas-boiler replacements resulting in an air-to-water heat pump in Belgium in 2030.
- $\widehat{RelExp}_i^{std}$ and $\widehat{RelShock}_i^{std}$ are the fitted values of the two standardized post-treatment beliefs obtained from their respective first-stage regressions.
- ε_i is the error term in the second-stage equation.

The causal interpretation of the 2SLS estimates assumes that the treatments affect heat-pump adoption through $RelExp_i$ and $RelShock_i$, rather than through other channels. A treatment may affect both beliefs, which is allowed by the first-stage specifications. Coibion et al. (2024) face the same identification issue: their information treatments affect several related expectations, and their causal interpretation likewise requires ruling out independent direct effects of the information on household decisions. In the present setting, the assumption would be violated if, for example, the treatments directly changed the attractiveness of heat pumps by making climate policy or environmental considerations more salient. The results are interpreted conditional on this assumption. Comparing the estimated belief-channel contributions with the intention-to-treat effects provides a useful diagnostic. Section 7.2 presents a robustness specification that allows each treatment arm to have an additive direct effect on adoption.

The primary comparison of the two mechanisms is a Wald test of

$$H_0 : \theta_1 = \theta_2. \quad (5.4.4)$$

The test uses the full 2SLS variance–covariance matrix. For each outcome, $\hat{\theta}_1$, $\hat{\theta}_2$, their estimated difference $\hat{\theta}_1 - \hat{\theta}_2$, the confidence interval for this difference, and the corresponding p -value are reported. Because standardized coefficients combine the effect per natural unit with the dispersion of the corresponding regressor, their comparison can depend on features of the study design and measurement process (Greenland et al., 1991). The Wald test is interpreted as comparing the effects of one-control-group-standard-deviation increases in the two beliefs. The standard-deviation scaling defines a common statistical benchmark for the comparison, but it does not make the two beliefs substantively equivalent. The benchmark depends on how each belief is elicited and on its dispersion in the control group. Following the distinction emphasized by Greenland et al. (1991), Section 6 therefore complements this coefficient comparison by evaluating the estimated contribution of the belief changes induced by the experimental interventions. This complementary analysis is explicitly specific to the numerical calibration of the treatments.

6 Complementary Analysis

6.1 Estimated Contributions of the Two Belief Channels

As a complementary analysis, the contribution of each belief channel to the effect of each treatment arm on heat-pump adoption is quantified using the belief changes actually induced by the numerical information provided in the experiment. This analysis is distinct from the primary comparison of standardized 2SLS coefficients: it combines the estimated effect of each belief with the magnitude of the corresponding belief shift generated by a particular treatment calibration.

For treatment arm k and adoption outcome m , let $\pi_{E,km}$ and $\pi_{S,km}$ denote the average first-stage effects on $RelExp_i$ and $RelShock_i$, expressed in their natural units. These effects are computed by averaging the predicted treatment–control differences across respondents, which incorporates the treatment-by-prior interactions in the first stages.

Let θ_E^{nat} and θ_S^{nat} denote the 2SLS effects expressed per natural unit of $RelExp_i$ and $RelShock_i$. They are obtained from the main standardized coefficients as

$$\theta_E^{nat} = \frac{\theta_1}{s_C(RelExp)}, \quad \theta_S^{nat} = \frac{\theta_2}{s_C(RelShock)}. \quad (6.1.1)$$

For outcome m and treatment arm k , the estimated contribution of each belief channel is defined as

$$\mathcal{C}_{E,km} = \theta_{E,m}^{nat} \pi_{E,km}, \quad \mathcal{C}_{S,km} = \theta_{S,m}^{nat} \pi_{S,km}, \quad (6.1.2)$$

and the combined contribution of the two belief channels as

$$\mathcal{C}_{km} = \mathcal{C}_{E,km} + \mathcal{C}_{S,km}. \quad (6.1.3)$$

These quantities are reported separately for each treatment arm and adoption outcome, together with confidence intervals obtained by bootstrapping the complete estimation procedure. The corresponding reduced-form intention-to-treat effect of each treatment arm on each outcome is also reported. Comparing $\mathcal{C}_{km}$ with that intention-to-treat effect assesses how much of the treatment-induced change in adoption is accounted for, within the model, by changes in $RelExp_i$ and $RelShock_i$. A close correspondence indicates that the two measured belief channels jointly account for most of the observed treatment effect, whereas a substantial gap indicates that the treatment effect is not fully captured by these channels. This comparison is a diagnostic of the two-channel linear model and is not imposed as an accounting identity.

The interpretation of this decomposition is deliberately treatment-specific. A larger contribution can arise either because a belief has a larger causal effect on adoption or because the numerical message used in the experiment produces a larger belief revision. The resulting contributions therefore describe the relative roles of the two channels under the experiment's chosen information values and should not be interpreted as evidence that one mechanism is generally more important outside this calibration.

6.2 Heterogeneous Treatment Effects by Risk Aversion

As a complementary reduced-form analysis, variation in the intention-to-treat effects on heat-pump adoption by pre-treatment risk aversion is examined. The lottery choice is coded from $RiskChoice_i = 1$ for Option A, the safest lottery, to $RiskChoice_i = 6$ for Option F, the riskiest lottery, with $RiskAversion_i = 6 - RiskChoice_i$. The resulting rank ranges from zero to five, with higher values indicating the choice of a safer lottery and therefore greater revealed risk aversion within the hypothetical task. Let $RiskAversion_i^c$ denote this rank centered on its mean in the full analysis sample. For each of the three adoption outcomes, the following specification is estimated:

$$\begin{aligned}
HeatPump_i = & \alpha + \sum_{j=1}^3 \beta_j T_{ji} + \phi RiskAversion_i^c + \sum_{j=1}^3 \omega_j (T_{ji} \times RiskAversion_i^c) \\
& + \delta Adopt_i^{pre} + \rho_1 RelExp_i^{pre} + \rho_2 RelShock_i^{pre} + \gamma' \mathbf{X}_i + \varepsilon_i.
\end{aligned} \tag{6.2.1}$$

Here, $\mathbf{X}_i$ contains the pre-treatment socio-demographic and dwelling controls other than risk aversion. Because risk aversion is centered, β_j is the intention-to-treat effect of treatment arm j for a respondent with the mean risk-aversion rank in the full analysis sample. The coefficient ω_j indicates how the treatment effect changes when the respondent moves by one rank towards a safer lottery. The primary heterogeneity test is $H_0 : \omega_2 = 0$ for the shock-exposure treatment, with $\omega_2 > 0$ expected: Treatment 2 should increase heat-pump adoption more strongly among respondents who choose safer lotteries. Because Treatment 3 also contains the shock-exposure information, $\omega_3 > 0$ is expected as a secondary directional prediction. No directional prediction is imposed on ω_1 . This reduced-form interaction captures overall moderation of the treatment effect and does not by itself distinguish whether risk aversion changes belief updating or the adoption response to a given belief change. Heteroskedasticity-robust standard errors are reported, and the specification is estimated separately for each adoption outcome.

7 Robustness

7.1 Alternative Measure of Relative Exposure to High Heating Costs

As a robustness analysis, respondents' expected frequency of shock-induced high-cost months between January 2030 and December 2035 is used. For each technology, respondents report the number of months, out of 72, in which they expect a major energy-price shock to cause the monthly heating cost to reach or exceed 120% of that technology's expected average 2030 cost. The alternative relative-exposure measure is defined as

$$RelShock_i^{months} = \frac{N_i(C_{2030-2035}^{gas} \geq 1.2E_i(C_{2030}^{gas})) - N_i(C_{2030-2035}^{HP} \geq 1.2E_i(C_{2030}^{HP}))}{72}. \tag{7.1.1}$$

The measure ranges from -1 to 1 . A positive value means that the respondent expects shock-induced high-cost months to be more frequent for gas heating than for the heat pump, while a negative value means the reverse. Unlike a ratio, the measure is defined when the respondent reports zero months for one or both technologies. $RelShock_i^{months}$ is standardized using its mean and standard deviation in the control group, and the treatment-effect and

2SLS specifications are re-estimated after replacing the main post-treatment $RelShock_i$ measure with $RelShock_i^{months}$. Because the months measure is elicited only after treatment, the pre-treatment probability-difference measure $RelShock_i^{pre}$ is retained as the baseline control in these robustness specifications.

7.2 Allowing for Direct Treatment Effects

To assess sensitivity to the exclusion restriction, the 2SLS model is re-estimated while allowing each treatment arm to have its own additive direct effect on heat-pump adoption. The robustness second stage is

$$HeatPump_i = \alpha + \theta_1 \widehat{RelExp}_i^{std} + \theta_2 \widehat{RelShock}_i^{std} + \sum_{j=1}^3 \tau_j T_{ji} + \rho_1 RelExp_i^{pre} + \rho_2 RelShock_i^{pre} + \delta Adopt_i^{pre} + \gamma' \mathbf{X}_i + \varepsilon_i. \quad (7.2.1)$$

In this specification, the treatment indicators are included directly in the adoption equation and therefore no longer serve as excluded instruments. The excluded instruments are the six treatment-by-prior interactions, $T_{ji} \times GapCost_i^{pre}$ and $T_{ji} \times RelShock_i^{pre}$ for $j = 1, 2, 3$. The first stages otherwise retain the same form as in equations (5.4.1) and (5.4.2). This specification allows each treatment to affect adoption directly by a constant amount, independently of the two measured beliefs. It still assumes that any such direct effect does not vary with respondents' pre-treatment beliefs. The two belief coefficients and their Wald comparison are reported as a robustness check; similarity with the main estimates is supportive of, but does not establish, the exclusion restriction.

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